From Four Rows to Five: Escape Times and Persistent Memory in Langton's Ant

Hao Ke

Abstract

How does one extra row change the time Langton's ant can spend inside a rectangle? For four rows and width n, the exact maximum first-exit time is $34n - 150$ for $n \geq 13$, with every smaller value also determined. A closed catalogue of 269 persistent boundary-response types gives this all-width result. Five rows instead admit infinitely many distinguishable boundary responses. An explicit pair of unary counters turns that memory into autonomous motion: a width-$4k + 9$ rectangle confines the ant for exactly $8k^2 + 60k + 16$ updates. This answers negatively the question of linear escape bounds at every fixed height. We then determine the entire future of the same two-counter family, including its earliest permanent period-104 highway entrance for every pair of counter lengths. Finally, an exact algorithm computes its complete state at any binary-specified time using fewer than 20,000 finite arithmetic XOR segments and polynomial bit time. Finite interface closure, local counter identities and integer-affine recovery certificates connect these results without resetting the field or discarding its history.

1 The exact question and answer

A finite rectangle contains only finitely many colourings, but enlarging its width creates exponentially many initial states. We seek the longest possible residence time for every width at once. Four rows provide a case where the answer is exactly linear and the effect of every return to a previously visited block can still be classified by a finite interface. The ant moves on $\mathbb{Z}^2$. At a white cell it turns right, changes the cell to black and moves forward by one unit; at a black cell it turns left, changes the cell to white and moves. Headings N,E,S,W are $0, 1, 2, 3$, with step vectors $(0, 1), (1, 0), (0, -1), (-1, 0)$, respectively. A state at time t is observed before update t. A pose consists of a position and a heading. For

$$R_n = \{0, \ldots, n-1\} \times \{0, 1, 2, 3\},$$

the first-exit time counts the update that moves the head out of R_n. The quantity $S_{4,n}$ is maximized over all 2^{4n} colourings, all $4n$ starting cells and all four headings. Colors outside R_n are arbitrary and remain unread and unchanged before this first exit. No boundary reflection or rule change is imposed.

Theorem 1. *The small values are*

n	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*
$S_{4,n}$	*2*	*18*	*28*	*46*	*64*	*86*	*104*	*146*	*156*	*206*	*224*	*266*

For every $n \geq 13$,

$$S_{4,n} = 34n - 150. \tag{1}$$

Every maximum is attained by a complete finite colouring and an effectively reconstructible internal pose.

Etse [1] asks whether $S_{4,n} \le 34n$ in Conjecture 11 of the published MFCS version, Section 4.3.3, p. 49:14. The adjoining text and Figure 13 give lower constructions of size $34n - 150$ for $n \ge 14$ and finite experimental agreement. The full arXiv version numbers this as Conjecture 13. Theorem 1 proves the conjectured upper bound and establishes the exact maximum for every width. Etse already supplied the lower-bound family and finite-width computations; the new step is a complete persistent-interface classification and an all-width upper certificate. Our attainment construction and small-width checks make the resulting exact classification reproducible. The coefficient 34 is consequently optimal. The first-exit convention agrees with [1]; our reduction includes all internal poses and does not rely on a boundary-start lemma.

Etse also asks whether a bound linear in width holds at every fixed height [1, Section 5, p. 49:16]. The five-row construction in Section 6 answers this question: it gives quadratic residence times and identifies the first height with infinite persistent response index. Section 7 follows the very same complete source through recovery to its permanent highway. Section 8 computes its whole state at arbitrary times from the two counter lengths, including the accumulated wake. Finite ant computation followed by a highway has precedents in the circuit construction of Gajardo, Moreira and Goles [2]; here the original counter source itself has an explicitly classified autonomous infinite future.

The four-row proof has three stages. Section 2 replaces each coloured block by the responses that a caller can observe after repeated entries, with all colour changes retained. Section 3 uses the finite response catalogue to bound local visits. Sections 4–5 attach durations and prove an exact recurrence by integer induction. Closure under adjoining a column gives coverage at every width; the weighted induction gives the exact escape time.

2 Persistent one-sided responses

A block is a contiguous sub-rectangle of four rows and any finite number of columns. Fix its near side, $s = 0$ for left or $s = 1$ for right. The other vertical boundary is its far side. A caller enters horizontally through the near boundary and observes the side and row at which the ant next leaves.

Put $\epsilon = (h - x - y) \bmod 2$. Each update changes both heading parity and cell parity, so ϵ is invariant. At column x write $p = (x + \epsilon) \bmod 2$; the parity of a block means that of its near column. On horizontal entry h is odd, so the entry row satisfies $y \equiv 1 - p \pmod 2$. There are exactly two such rows,

$$I_p = (1 - p, 3 - p).$$

A horizontal exit enters the neighbouring column of parity $1-p$, so its row lies in $U_p = I_{1-p} = (p, p+2)$. Each list has channels $0, 1$ in the order displayed. A four-cell column has mask $b = \sum_{y=0}^{3} \mathbf{1}_{\text{black}}(y) 2^y$.

A caller may repeatedly choose one of the two incoming channels at this side. Each request initiates a literal excursion until the head exits the block. A near-side exit returns its channel and the behaviour of the entire resulting colouring. A top, bottom or far-side exit is terminal. The colouring is never reset between requests.

Definition 2. A finite response type is an inductively generated finite tree represented by a pair $t = (\rho_0, \rho_1)$, where $\rho_i = \perp$ or $\rho_i = (j, t')$, with $j \in \{0, 1\}$ and t' a finite response type. Its rank is

$$r(t) = \max\big(\{0\} \cup \{1 + r(t') : \rho_i = (j, t')\}\big).$$

The empty block has type $t_0 = (\perp, \perp)$.

We do not assume in advance that every physical block has a finite response type. This will follow from a checked closed algebra. A type forgets the duration and the distinction among terminal sides, since there can be no subsequent request after a terminal. It retains every channel and residual response after a successful return. It is observational equivalence for this interface: different physical colourings have the same type precisely when their future interface responses agree. Durations are reintroduced in Section 4.

2.1 Adjoining a column

Suppose a near column has mask b, parity p, and a child block has near side s, parity $1-p$, and type r. Denote the combined type by $\Phi_{s,p}(b,r)$. A literal one-column excursion takes at most two updates: a horizontal incoming heading makes a vertical move and then a horizontal move, unless a vertical boundary has already been crossed; a vertical incoming heading leaves horizontally in one update. Thus its transition table is an exact finite table.

For a parent request execute this one-column table. A vertical exit terminates. A near exit returns its channel, with residual product $\Phi_{s,p}(b',r')$. A far exit enters the child at its corresponding channel. A terminal child answer terminates the parent. Otherwise retain both the changed column mask and the child's returned residual type, reenter the column from its far side and continue. Each successful child call lowers its rank. A proposed finite product table is checked against these equations; on a parent return, the resulting type must be exactly the declared residual product and have smaller rank than the parent type.

Proposition 3 (Finite closure certificate)**.** *For each $(s,p)\in\{0,1\}^2$ there are 269 distinct canonical response types. They include t_0, have ranks at most 13, and are closed under all 16 column masks and all 269 child types. All $4\cdot 16\cdot 269 = 17,216$ product cases are verified for both input channels. The realized type sets by width have sizes*

width	*0*	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*
size	*1*	*12*	*36*	*83*	*158*	*240*	*264*	*269*	*269*

The sets at widths seven and eight are equal, not merely equinumerous. Every catalogue type has a physical representative of width at most seven.

Proof and infinite-width interpretation. The finite facts are checked by the supplied reconstruction and independent verifier. Nodes are canonical pairs of terminal or channel/child responses; acyclicity and ranks are checked. Product equations are verified by literal one-column updates, preserving the child's residual state. For coverage, induct on width. The empty block has t_0. Every wider colouring is exactly a near column and a smaller coloured block. By the induction hypothesis the latter has its exact response type. The checked equations therefore simulate the real parent and establish its exact residual type. Ranks make every request and every permitted series of returns finite. The closed table ensures the result stays in the catalogue for every width. The displayed representatives, independently replayed as complete response trees, prove realizability. Distinct canonical trees have some finite request sequence distinguishing them. Thus the catalogue is exact for the stated interface. □

Writing $\mathcal{T}_w^{s,p}$ for the realized types at width w, composition gives

$$\mathcal{T}_{w+1}^{s,p} = \{\Phi_{s,p}(b,r) : 0\le b<16,\ r\in\mathcal{T}_w^{s,1-p}\}.$$

Equality of the width-seven and width-eight sets for both parities therefore implies stabilization at every larger width. Each block permits at most 13 successful returns, and this rank is attained by a compatible request sequence.

3 A local bound independent of width

Consider a column at its first visit during a rectangle-confined run. Its left and right parts have arbitrary complete colourings, possibly modified by earlier motion, and hence have exact catalogue types. Replace an excursion into either side by its persistent response. A return lowers the sum of the two ranks. Between responses there are at most two central-column updates. This product terminates without any assumption on the widths of the side blocks.

Proposition 4 (Sharp column and cut bounds)**.** *Before first escape from any four-row rectangle, a fixed column is updated at most 34 times. A vertical cut is crossed at most 21 times. Both bounds are attained.*

Proof. The central-column check enumerates the two parities, all pairs of 269 side types, 16 masks, and eight compatible internal row/heading poses. For each of the four rows, $h - y \equiv p \pmod 2$ allows two headings, giving exactly $2 \cdot 269^2 \cdot 16 \cdot 8 = 18,524,416$ cases. The largest local count is 34. The cut handshake chooses two parities, two entry sides, two entry channels and a pair of block types: $2 \cdot 2 \cdot 2 \cdot 269^2 = 578,888$ cases, counting the first entry crossing; its largest count is 21. Each internal return lowers a checked rank, so these are exhaustive finite products, not truncated simulations. For sharpness of the column bound, the masks

$$(15, 9, 9, 9, 7, 1, 11, 6, 6, 6, 0)$$

with head $(5, 1)$ facing west give 159 total updates and exactly 34 in column 5. For the cut bound, masks $(11, 6, 6, 2, 13, 12, 4, 12, 6, 13)$ with head $(4, 2)$ facing south give 164 updates and 21 crossings between columns 4 and 5. Both witnesses are autonomous runs with their complete initial colourings. □

Summing column counts proves $S_{4,n} \le 34n$ immediately. This already resolves the published conjecture. It does not yet give the subtractive constant 150, because maximal column behaviours need not be simultaneously attainable. The next reduction addresses precisely that compatibility issue.

4 An exact weighted recurrence

The duration of a block cannot be maximized independently on each call: the same colouring must persist across all of them. We retain the whole finite request script in a weighted query. We construct rectangles by adjoining columns on the left, so all weighted queries use near side $s = 0$. A script records the incoming channels; the response type determines all returned channels and residual types along that script.

An *ordinary query* $F(p, t, q)$ specifies parity p, response type t and a finite channel word $q \in \{0, 1\}^*$. All requests except possibly its last must return; the last may return or terminate. Write $V_n(F(p, t, q))$ for the maximum total updates of this script over all width-n colourings of type t. If no such colouring exists, the value is $-\infty$. The empty script has zero cost only when its type is

realizable at that width. Rank at most 13 permits at most thirteen successful returns, optionally followed by one terminal request. Relevant scripts therefore have at most fourteen requests; the same bound applies to the child scripts generated by the decomposition below.

A *rooted descriptor* o is terminal or a pair (j, t): the first run starts somewhere inside the block and either terminates or returns at left-side channel j, leaving ordinary type t. There are $1 + 2 \cdot 269 = 539$ descriptors per parity. A rooted query $R(p, o, q)$ performs this initial run followed by q. If o is terminal, q is empty. Its value maximizes over all complete colourings and all internal starting poses realizing that descriptor.

4.1 Complete decomposition rules

For an ordinary query enumerate every (b, r) with $\Phi_{0,p}(b, r) = t$. Run the parent script symbolically through that column and its child's exact response tree. The result is a local count c and a child script q'. This is an edge to $F(1 - p, r, q')$ of weight c.

For a rooted query there are two, and only two, possible locations for the root. If it is in the new column, enumerate all eight compatible poses, the mask and the child's ordinary type, run the initial excursion, then the parent script. The successor query is ordinary F. If the root is in the child, enumerate its rooted descriptor. A terminal child first run gives a terminal parent without touching the new column. Otherwise its output channel starts the new-column motion. The later calls form the continuation script of a rooted child query R. In both cases count only updates in the new column in c; the child's query counts its entire history exactly once.

The starting goals are all rooted descriptors with empty continuation. Closing these rules backwards yields 21,248 states and 527,764 edges, with 1,078 goals. The graph is finite because the catalogue and bounded scripts are finite. The verifier checks every required successor and edge, both root placements, and reachability of every stored state from the goal set.

Use $V_n(v)$ for the value of either kind of query. The recurrence is

$$V_0(v) = \begin{cases} 0, & v = F(p, t_0, q),\ |q| \le 1, \\ -\infty, & \text{otherwise}, \end{cases}$$
$$V_n(v) = \max_{v \xrightarrow{c} u} \bigl(c + V_{n-1}(u)\bigr). \tag{2}$$

An empty block terminates its first ordinary request without an update and has no internal starting root. These are exactly the stated base cases.

Lemma 5 (Physical exactness)**.** *The values in* (2) *equal the physical maxima of their queries. Every finite value is attained. Consequently*

$$S_{4,n} = \max_{v \in \mathcal{G}} V_n(v), \tag{3}$$

where $\mathcal{G}$ is the set of all 1,078 goals.

Proof. Induct on width. A real colouring splits into its near-column mask and its actual child. For rooted queries the initial head is either in that column or in the child. These exhaustive alternatives select one enumerated edge. Persistent response equivalence forces the recorded child script, so the time is local weight plus one child query value. This proves the upper inequality. Conversely, take an attaining child configuration for a finite successor value and prepend the edge's recorded column.

If appropriate, use its recorded initial pose; otherwise retain the child's root. The exact response equations ensure that all successive calls are made to this same child with its evolving colours. They give the parent duration and descriptor claimed by the edge. This proves attainment and the lower inequality. Maximizing over all rooted outcomes gives (3). No boundary-entry assumption is used. □

Rooted queries supply the internal starting poses, while whole-script queries preserve compatibility across repeated entries into the same child.

5 All-width affine stabilization by local integer inequalities

The recurrence through width 23 is computed exactly. Define integer vectors on all 21,248 states by

$$A = V_{20}, \quad B = V_{21}, \quad d = V_{22} - V_{20}, \quad e = V_{23} - V_{21}.$$

The certificate checks that A and B are finite everywhere. On every edge $v \xrightarrow{c} u$ it verifies

$$B(v) \geq c + A(u), \qquad e(v) \geq d(u), \tag{4}$$

$$A(v) + d(v) \geq c + B(u), \qquad d(v) \geq e(u). \tag{5}$$

For each vertex there is an edge attaining both inequalities in (4), and an edge, possibly different, attaining both in (5). There are 2,111,056 integer inequalities and 42,496 tight-edge existence checks.

Lemma 6 (Explicit affine certificate)**.** *These checked inequalities and tight edges imply, for every* $k \geq 0$*,*

$$V_{20+2k} = A + kd, \qquad V_{21+2k} = B + ke.$$

Proof. Substitution into (2) gives $B + ke$ as an upper bound by (4); a simultaneously tight edge attains it. Substituting once more, (5) gives and attains $A + (k+1)d$. The base values are the exact computed V_{20}, V_{21}. This proves the assertion by induction, not by an empirical period-detection rule. □

For every goal the checked envelopes are

$$A \leq 530, \quad B \leq 564, \quad d \leq 68, \quad e \leq 68.$$

There is a goal with $A = 530, d = 68$ and a goal with $B = 564, e = 68$. Lemma 6 and (3) give $34n - 150$ for all $n \geq 20$. The goal maxima computed directly at widths 13 through 19 already have this affine value, before the full state-vector induction starts at width 20. These values and the twelve smaller widths prove Theorem 1.

This is also a constructive lower-bound proof. At each width choose an attaining edge, output its near-column mask and recurse on the child. Exactly one rooted to ordinary transition supplies the initial head. The path ends at a valid empty block base case. Thus every n yields a complete maximizing seed, with n explicit masks. Its reconstruction is an output-sensitive finite algorithm, not a demand to simulate an unbounded run. In particular the asymptotic lower coefficient 34 is established independently of the earlier lower construction.

6 Five rows and autonomous persistent memory

Let $S(h, n)$ be the largest first-exit time from an h-row, n-column rectangle, over all initial colors and all internal poses. The exit-producing update is counted. Exterior cells are never read before this exit and may be arbitrary; our constructions set them white.

Theorem 7 (Quadratic five-row confinement)**.** *For every $k \in \mathbb{N}$, an explicitly given complete finite coloring has $10k + 23$ black cells in a five-row rectangle of width $n = 4k + 9$. From its specified pose, the first-exit time is*

$$T_k = 8k^2 + 60k + 16 = \tfrac{1}{2}n^2 + 6n - \tfrac{157}{2}.$$

The trajectory crosses its central vertical seam exactly $2k$ times. Its complete coloring after each round and at exit is given explicitly. Consequently $S(h, n) = \Omega(n^2)$ for each fixed $h \geq 5$.

This is an exact formula for the constructed family, not for the maximum $S(5, n)$. It answers the concluding fixed-height linear-bound question of [1], while retaining that paper's polynomial upper bound. Together with the linear bounds through height four [1] and Theorem 1, five is the least height admitting superlinear worst-case first escape.

A second result concerns a deliberately modest observable. A block is repeatedly entered from the same vertical side. Colors persist between calls. We observe the returned row, or termination when the ant leaves another side, but not elapsed time or which terminal side is used.

Theorem 8 (Boundary-response threshold)**.** *For each fixed entry side and parity sector, this interface has respectively $1, 2, 7, 269$ response types at heights $1, 2, 3, 4$, over all finite widths and all colorings. At every height $h \geq 5$ its index is infinite. More precisely, at width $2m + 4$ there are $m + 1$ pairwise distinguishable five-row colorings, even when the only experiment is repetition of a single input row.*

The height-four catalogue was established above; the same finite closure check gives the smaller-height cases. The construction below establishes the infinite-index side and its autonomous composition.

6.1 Complete column states and local transformations

6.1.1 Conventions

Rows are $y = 0, 1, 2, 3, 4$. Headings north, east, south, west are $0, 1, 2, 3$. Time t is observed before update t. A column mask is $b = \sum_{y=0}^{4} b_y 2^y$, with $b_y = 1$ for black. A word of masks denotes all colors in consecutive columns, left to right. Products concatenate; powers are ordinary finite repetitions, not symmetric differences.

Define four fixed complete column words

$$A = (6, 5), \qquad B = (10, 9), \qquad C = (6, 10, 12, 17), \qquad V = (18, 29), \tag{6}$$

and, for $j, k \geq 0$,

$$\mathcal{R}(j, k) = A^j C V^k, \qquad \mathcal{L}(j, k) = (5)\mathcal{R}(j, k). \tag{7}$$

The respective entry poses are $(0, 3), \mathrm{E}$ and $(0, 2), \mathrm{E}$. Every cell outside a displayed block is white unless part of an explicitly specified neighboring block. An excursion ends as soon as its head leaves the block, before reading an exterior cell.

6.1.2 Six finite kernels

The following table gives literal executions. Every pre-update read occurs inside the displayed block; the final move is the only block exit.

Input	Entry	Turns	Output	Steps	Exit
A	$(0,3),\mathrm{E}$	P	B	4	$(2,3),\mathrm{E}$
B	$(1,2),\mathrm{W}$	P	A	4	$(-1,2),\mathrm{W}$
CV	$(0,3),\mathrm{E}$	K	AC	30	$(-1,2),\mathrm{W}$
C	$(0,3),\mathrm{E}$	E	$B(10,18)$	16	$(4,1),\mathrm{E}$
(5)	$(0,2),\mathrm{E}$	LR	(9)	2	$(1,3),\mathrm{E}$
(9)	$(0,2),\mathrm{W}$	RL	(5)	2	$(-1,3),\mathrm{W}$

The turn words are

$$P = \texttt{RLLR},$$
$$K = \texttt{RLRLRRRRLLLLRLLRLRLLRLLRLRRLLR},$$
$$E = \texttt{RLRLRRRRLLLLRLLR}.$$

These are finite proof obligations, not hypotheses inferred from long simulations. For example, the reads on A are $(0,3),(0,2),(1,2),(1,3)$, requesting R, L, L, R and changing the masks to $(10,9)$. On B the westward reads are $(1,2),(1,3),(0,3),(0,2)$ and restore $(6,5)$. For CV and C, the displayed words, initial masks, and pose determine all 30 or 16 reads; direct application of the rule verifies the output masks and exit. The supplementary immutable witness lists each read coordinate and the complete output for all six rows.

6.2 A unary counter with a persistent complete residual

Lemma 9 (Counter recurrence)**.** *For $k > 0$, an excursion in $\mathcal{R}(j,k)$ from $(0,3),\mathrm{E}$ returns at $(-1,2),\mathrm{W}$ after $8j+30$ updates and leaves exactly $\mathcal{R}(j+1,k-1)$. For $k=0$, it exits at $(2j+4,1),\mathrm{E}$ after $4j+16$ updates, leaving $B^{j+1}(10,18)$.*

Proof. Apply the eastward $A \to B$ kernel j times. The prefix A^j becomes B^j and the head reaches the first column of C, at row 3 facing east, after $4j$ updates. No later block has yet been read.

When $k > 0$, apply the $CV \to AC$ kernel. Its final position is the preceding column at row 2 facing west. It then traverses B^j backwards, using the $B \to A$ kernel j times. The prefix is restored, the shifted cap is AC, and the untouched suffix is V^{k-1}. Thus the entire block is $A^{j+1}CV^{k-1}$ and the duration is $4j + 30 + 4j$. All stages read only their stated subblocks; a final move into a neighboring block does not update it.

When $k = 0$, the terminal C kernel exits right. There is no return scan, so B^j remains and the complete output is $B^{j+1}(10,18)$. The duration is $4j + 16$. None of the kernels exits through a horizontal boundary. □

Consequently, repeated row-3 requests produce k row-2 replies followed by a terminal reply on request $k+1$. The residual after q successes is exactly $\mathcal{R}(j+q,k-q)$, not a reset or a retained core with unspecified debris.

Lemma 10 (Port adjustment)**.** *A successful request to $\mathcal{L}(j,k)$ from $(0,2),\mathrm{E}$, for $k>0$, returns at $(-1,3),\mathrm{W}$, takes $8j+34$ updates, and leaves $\mathcal{L}(j+1,k-1)$. At $k=0$ it exits far at $(2j+5,1),\mathrm{E}$ in $4j+18$ updates, leaving $(9)B^{j+1}(10,18)$.*

Proof. The leading column executes $5\to 9$ on entry, supplying the row-3 entry for Lemma 9. A successful return supplies row 2 facing west to the $9\to 5$ kernel, restoring the leading column and returning at row 3. At exhaustion there is no return through that column, so its mask remains 9. □

Proposition 11 (Distinguishable histories at a common width)**.** *For fixed $m\geq 0$, the complete colorings $\mathcal{R}(m-k,k)$, $0\leq k\leq m$, all have width $2m+4$ and have pairwise distinct time-blind same-side response types.*

Proof. For two remaining counts $k<k'$, issue $k+1$ copies of the row-3 request. The first block terminates on the last request while the second still returns row 2. Both share the same physical width; neither time nor interior colors are observed. Adding white rows above row 4 does not change any read, and termination is through the far vertical boundary. The construction therefore works at every height at least five. Lemma 10 covers the other entry parity, and rotation covers the opposite side. □

An exact summary capable of answering these experiments at width $2m+4$ must therefore have at least $m+1$ values, or at least $\lceil\log_2(m+1)\rceil$ bits. This is a memory lower bound for the specified response observable as the finite source width grows.

6.3 Autonomous composition and the exact quadratic exit

An interface request is an intervention: the experimenter supplies the next entry. We now remove that intervention entirely.

Let $\mathcal{J}$ reflect a five-row block in the line $x=-1/2$ and complement every color inside that block. Reflection alone exchanges left and right turns; color complementation exchanges their conditions, restoring the original rule. Thus $\mathcal{J}$ is an exact conjugacy on every update inside the block. This is only a local transformation up to exit: the infinite exterior is not complemented and the constructed black set remains finite.

For $l,r\geq 0$, place $\mathcal{R}(0,r)$ in columns $0,\dots,2r+3$, and the reflected, complemented $\mathcal{L}(0,l)$ in columns $-2l-5,\dots,-1$. Start at $(0,3),\mathrm{E}$. In an ordinary left-to-right list the complete input is

$$\mathrm{rev}\big(31-((5)CV^{l})\big)\;CV^{r}, \tag{8}$$

where $31-$ acts entrywise. The rectangle has width $2(l+r)+9$ and exactly $6r+4l+23$ black cells. The rest of the plane is white.

A right return has pose $(-1,2),\mathrm{W}$, precisely the entry for the reflected left block. A left return has pose $(0,3),\mathrm{E}$, precisely the right entry. No pointer, phase, color, or environment is reset between requests.

Lemma 12 (Complete round induction)**.** *After j complete right/left pairs, for $0\leq j\leq\min(l,r)$, the head is $(0,3),\mathrm{E}$ and the full two-sided field is*

$$\mathcal{J}\big[(5)\mathcal{R}(j,l-j)\big]\quad|\quad\mathcal{R}(j,r-j). \tag{9}$$

The time is $t_j=8j^2+56j$. No prior update has left the original rectangle.

Proof. The claim at zero is the definition. If both counters are positive, the next right excursion takes $8j + 30$ steps and the next left excursion takes $8j + 34$, by Lemmas 9 and 10 with the local conjugacy. Their complete outputs give (9) at $j + 1$. Their kernel footprints exclude premature vertical or far exits. Summing $16j + 64$ over $j = 0, \dots, q - 1$ gives $8q^2 + 56q$. □

Theorem 13 (Two-parameter complete exit)**.** *The source* (8) *first exits at*

$$T(l, r) = \begin{cases} 8r^2 + 60r + 16, & r \le l, \\ 8l^2 + 68l + 48, & l < r. \end{cases}$$

For $r \le l$, the exit pose is $(2r + 4, 1), \mathrm{E}$, and the full endpoint is

$$\mathcal{J}[(5)\mathcal{R}(r, l - r)] \quad | \quad B^{r+1}(10, 18).$$

For $l < r$, the exit pose is $(-2l - 6, 1), \mathrm{W}$, and the full endpoint is

$$\mathcal{J}[(9)B^{l+1}(10, 18)] \quad | \quad \mathcal{R}(l + 1, r - l - 1).$$

Proof. If $r \le l$, execute r full pairs and then the right terminal call. Its $4r + 16$ additional updates give the first formula and the full endpoint. If $l < r$, execute l pairs, one successful right call costing $8l + 30$, and a terminal left call costing $4l + 18$. These give the second formula and endpoint. The footprint guards prove this is the first exit, not only an observed exit. □

Taking $l = r = k$ proves Theorem 7. Each completed excursion crosses the seam once; the terminal right excursion does not cross it, so there are $2k$ seam crossings. The pair-checkpoint black mass stays $6r + 4l + 23$: the right block loses two black cells per pair while the complemented left block gains two. Equality of black mass and pose does not mean equality of complete states; the cap locations and rewritten columns change.

For arbitrary $n \ge 9$, use $k = \lfloor (n - 9)/4 \rfloor$ and add white columns to the observation rectangle. Enlarging a region cannot decrease its first-exit time. White rows added above likewise preserve the lower bound. Hence no finite-valued $f(h)$ can bound $S(h, n)$ by $f(h)n$ for all h, n. In particular, the open linear-bound question of [1] already has a negative answer at height five. The known $O(n^3)$ upper bound there leaves a gap; we do not claim optimality of the quadratic coefficient or of the constructed family.

6.4 The finite/infinite response boundary

To complete Theorem 8, we specify the finite closure being used on the other side of the threshold. The invariant $\epsilon = (h - x - y) \bmod 2$ fixes the horizontal entry rows of a column. Writing $p = (x + \epsilon) \bmod 2$, entry rows have parity $1 - p$ and output rows have parity p.

A finite persistent response type is a tuple with one branch for each allowed input row. Each branch is terminal or consists of an output channel and a residual type. Define its rank as the maximal number of successful returns along any input sequence. Colors persist through all these calls. Terminal side and time are deliberately unobserved, so this quotient does not by itself encode the exact time calculations in the preceding sections.

6.4.1 Finite certificate and its interpretation

The finite catalogs are

Height	Types per side/parity	Maximal rank	Closure width
1	1	0	1
2	2	1	2
3	7	3	3
4	269	13	8

The fourth row is the persistent response classification of Proposition 3. The supplementary checker independently rebuilds all four rows. For height four the exact-width type sets, at widths zero through eight, have sizes

$$1,\ 12,\ 36,\ 83,\ 158,\ 240,\ 264,\ 269,\ 269.$$

Importantly, the last two *sets*, not merely their cardinalities, coincide.

Adjoin any possible near-column mask to any opposite-parity child type. A horizontal one-column excursion makes at most two updates, then exits that column or the strip. If it enters the child, use the child's declared response, retain its residual type and resume with the modified near-column mask. Every successful child reply decreases child rank. A near-side parent reply must match the declared residual product and decrease parent rank.

The certificate checks these local equations for every mask, child type and input channel. Induction on width proves their physical interpretation: assuming the complete child has the specified persistent behavior, the exact one-column transitions and its retained replies uniquely determine the parent. Child rank ensures each request terminates; parent rank controls repeated requests. The empty block starts the induction. Every catalog type has a concrete representative, whose full response tree is also replayed.

Finally, distinct finite labeled response trees have a distinguishing finite input word: select a branch on which they differ, or recurse to distinct residuals. Thus the catalog is an exact quotient for the stated observable, not only an overapproximation. Proposition 11 proves infinite index starting at five rows and completes Theorem 8.

6.5 Exact macro-dynamics beyond semilinear output

The counter construction admits a compact macro-description: the processed length j, remaining lengths $l-j, r-j$, and current side specify the complete column words and pose. Counter updates and the duration of one call are affine. There are, however, an input-dependent number of calls, and their sum is quadratic.

Corollary 14. *There is a Presburger-definable finite-fiber family of complete initial ant states whose exact first-exit-time graph is not semilinear.*

Proof. In (8) with $l = r = k$, black membership is specified by linear interval endpoints, parity and fixed row masks. Hence the source family and its observation rectangle are Presburger-definable. Its time graph is $\{(k, 8k^2 + 60k + 16) : k \geq 0\}$.

If this graph were a finite union of linear sets, some infinite component would contain $(k_0 + ma, t_0 + mb)$ for all $m \geq 0$, with nonzero period (a, b). A graph of a function cannot have

$a = 0, b \neq 0$; nonnegativity forces $a > 0$. The quadratic identity then has a nonzero m^2 coefficient $8a^2$, contradicting the affine second coordinate. Thus the graph is not semilinear. □

Thus a finite-parameter macro-description can give a nonsemilinear accumulated time graph. The displayed quadratic remains directly evaluable in polynomial bit time, a distinction used in the complete-state algorithm below.

7 The complete two-counter highway basin

A standard highway is permanent agreement of the pose and turn trace with the usual translating period-104 word, up to phase and quarter-turn rotation. Historical black cells need not become a translated finite motif. To fix the convention completely, our standard word is the concatenation of these two displayed strings:

```
RRLLLLRLRRRRLRRRRLLLLRLRRRRLRLLRRLLLLRRLLRRRRLLRRLRL
LRLLRLRLLRRRRLLRLLRRRRLLRRRRLLRLRRRRLRLLLLRRRRLRRLRR
```

It has primitive length 104; its cyclic phases and quarter-turns are allowed.

For the remainder of the paper we translate the source of Section 6 so its leftmost column is $x = 0$. This changes its initial head from $(0, 3)$ to $(2l + 5, 3)$ and leaves all durations unchanged.

7.1 Encoding

A column of height five is the integer $\sum_{y=0}^{4} b_y 2^y$. Column words are placed from $x = 0$ to the right in rows $0, \ldots, 4$; every omitted lattice cell is white. Set

$$A = (6, 5), \quad B = (10, 9), \quad C = (6, 10, 12, 17), \quad V = (18, 29), \qquad \mathcal{R}(j, k) = A^j C V^k.$$

For a column word define

$$\mathcal{J}(c_1, \ldots, c_n) = (31 - c_n, \ldots, 31 - c_1).$$

Thus $\mathcal{J}$ reflects and complements *inside the indicated finite block*; it does not complement the exterior. For $l, r \in \mathbb{N} = \{0, 1, \ldots\}$, let $X(l, r)$ have column word and initial pose

$$\mathcal{J}((5)CV^l)\, CV^r, \qquad ((2l + 5, 3), \mathrm{E}). \tag{10}$$

Its width is $n = 2(l + r) + 9$, its initial black mass is $4l + 6r + 23$, and all these cells are retained throughout our argument.

Theorem 15 (Complete counter basin)**.** *For every $l, r \in \mathbb{N}$, the source $X(l, r)$ eventually enters the standard highway. Its earliest permanent entrance time is*

$$\tau(l, r) = E(l, r) + \rho(l, r), \qquad E(l, r) = \begin{cases} 8r^2 + 60r + 16, & r \leq l, \\ 8l^2 + 68l + 48, & l < r, \end{cases}$$

where E is the first rectangle-exit time and ρ is completely specified by Tables 1 and 2. In the balanced case,

$$\tau(k, k) = 8k^2 + 100k + 1047 \quad (k \geq 2), \tag{11}$$

while $\tau(0, 0) = 11576$ and $\tau(1, 1) = 1024$.

Region	Condition	Recovery delay ρ
$r \le l$, $d = l - r$	$d = 0, r = 0$	11560
	$d = 0, r = 1$	940
	$d = 0, r \ge 2$	$40r + 1031$
	$d = 1$	$32r + 883$
	$d \ge 2$	$32r + 799$
$l < r$, $l \ge 9$, $d = r - l$	$d = 1$	$72l + 1427$
	$d = 2$	$128l + 2611$
	$d = 3$	$144l + 3011$
	$d = 4$	$176l + 3658$
	$d = 5$	$176l + 3678$
	$d \ge 6$	$176l + 3570$

Table 1: All unbounded parameter cones. The remaining fixed-l strips are covered by Table 2, not omitted.

l	Delays for $d = r - l = 1, 2, \ldots$; the final entry repeats forever
0	$17106, 2789, 1479, 2661, 5711, 2665, 2557$
1	$1516, 1423, 1591, 6410, 6430, 6322$
2	$1547, 1925, 1945, 1837$
3	$1635, 2021, 2041, 1933$
4	$1715, 2883, 3787, 3915, 3413, 3433, 3325$
5	$1787, 3071, 3447, 4199, 4495, 4859, 6975, 7638, 7658, 7550$
6	$1859, 4607, 4451, 4471, 4363$
7	$1931, 3543, 3931, 4491, 4771, 4470, 4490, 4382$
8	$2003, 3635, 4163, 6057, 6077, 5969$

Table 2: The finite base table and its protected infinite extensions. Each row is a formula for all $d \ge 1$, not an observation up to its printed length.

The theorem is computer-assisted in a precise finite sense. The infinite domains are checked by integer-affine identities with proved coverage, and only fifty-nine isolated states require literal finite execution. We supply full words, source formulas and exact verification programs. None of the parameter cutoffs in the finite controls is a premise of the theorem.

The round induction and two terminal kernels in Section 6, after the indicated translation, give the complete exit states

$$\begin{aligned} r \le l: &\quad \mathcal{J}((5)\mathcal{R}(r, l-r))B^{r+1}(10, 18), \quad (n, 1), \mathrm{E}, \\ l < r: &\quad \mathcal{J}((9)B^{l+1}(10, 18))\mathcal{R}(l+1, r-l-1), \quad (-1, 1), \mathrm{W}. \end{aligned} \tag{12}$$

7.2 Recovery certificates for every parameter

7.2.1 The nine affine-word programs

A program is a finite concatenation

$$u_0 v_1^{a_1 k + b_1} u_1 \cdots v_s^{a_s k + b_s} u_s, \tag{13}$$

Program	(l, r)	Domain	Length	Powers	Read tests
E0	(k, k)	$k \geq 2$	$40k + 1031$	6	1007
E1	$(k+1, k)$	$k \geq 0$	$32k + 883$	4	839
E2	$(k+2, k)$	$k \geq 0$	$32k + 799$	4	751
W1	$(k, k+1)$	$k \geq 9$	$72k + 1427$	9	1431
W2	$(k, k+2)$	$k \geq 9$	$128k + 2611$	17	2795
W3	$(k, k+3)$	$k \geq 9$	$144k + 3011$	19	3203
W4	$(k, k+4)$	$k \geq 9$	$176k + 3658$	22	4050
W5	$(k, k+5)$	$k \geq 9$	$176k + 3678$	22	4074
W6	$(k, k+6)$	$k \geq 9$	$176k + 3570$	22	3970

Table 3: A test for a variable phase covers all its repetitions, not one sample. The sum is 22,120 read obligations on unbounded domains.

where the u_i, v_i are explicit LR words. Each variable block v_i restores the heading and has displacement $(2, 0)$ or $(-2, 0)$ in its actual pose. The counts are positive on the declared entire ray $k \geq k_0$. Table 3 gives the domains and sizes. The literal fixed words and integer exponents are the immutable data in `witness.json`.

For illustration, E0 has fixed-block lengths $114, 108, 3, 9, 3, 169, 577$, separated by powers of

$$\mathtt{RLLR}^{2k+3}, \quad \mathtt{LRRL}^{2k+3}, \quad \mathtt{RRLL}^{2k+3},$$
$$\mathtt{LRRL}^{k+1}, \quad \mathtt{RRLL}^{k+1}, \quad \mathtt{LRRL}^{2k+1}.$$

The length is $40k + 1031$. This word pattern is not accepted merely because finite runs exhibit it: every read is proved correct by the following algebra.

7.2.2 Complete affine fields

Throughout a program the row of a phase is independent of k and its horizontal position is $ak + b$, with a even. Variable repetitions add $\pm 2j$. Fix a row y and horizontal parity $p \in \{0, 1\}$, and write $x = 2z + p$. A black arithmetic interval is represented exactly by two jumps:

$$[L(k) \leq z \leq U(k)] = [z \geq L(k)] \oplus [z \geq U(k) + 1]. \tag{14}$$

Here L, U have integer affine coefficients. Equal jumps cancel by XOR. Zero-length intervals cancel as well. This representation includes all initial cells and all complete earlier writes; it never extracts an assumed active subset.

Suppose the phase reads $q_i + jd$ in a variable block, with $d = (\pm 2, 0)$. The pre-block field is an XOR of tests $j \geq f(k)$ for integer-affine f. For a previous phase q_t, the unique copy that could have written this cell has index $j - s$, where $q_t - q_i = sd$. It is earlier exactly when $s > 0$, or $s = 0$ and $t < i$, and it exists exactly when $j \geq s$. Include those constant thresholds and the required L bit at $j = 0$. The read condition becomes

$$\bigoplus_{f \in \mathcal{E}} [j \geq f(k)] = 0, \qquad k \geq k_0, \quad 0 \leq j < ak + b. \tag{15}$$

Fixed reads are the special case with one epoch. Once a block is verified, all of its phase intervals are added to the complete field. Induction over ordinary time proves that satisfying these constraints executes the literal program, rather than just an abstract trace.

Lemma 16 (Exact affine epoch test)**.** *For finite $\mathcal{E}$ and a positive affine epoch count on $k \geq k_0$, condition (15) is decidable by finite integer sign and intersection checks, including the final infinite parameter interval.*

Proof. First check parity at $j = 0$. It changes only at the roots $f(k) = 0$; isolating $\lfloor \text{root} \rfloor$ and its successor separates all integer sign regions. For every distinct affine jump f, solve $1 \leq f(k) \leq ak+b-1$. This is an integer interval. If it is infinite, the jump cannot coincide with other distinct affine jumps throughout it: finitely many other lines give only finitely many intersections. Thus the condition fails. If the interval is finite, enumerate every integer intersection with other jumps. Their distinct parameter values must cover the whole interval, and every coincident jump there must have even multiplicity. These conditions are exactly absence of odd interior jumps. Along with zero initial parity they prove the claimed XOR identity. □

A second implementation instead partitions at every pairwise affine-order root, including comparisons with zero and the epoch count. In each resulting integer interval all order relations are fixed, so one representative checks initial parity and every interior multiplicity. Both algorithms use exact integers; neither uses floating-point feasibility, a parameter cutoff, or an SMT solver. They agree on the complete fields and lengths of all nine programs. The common geometric compilation remains an explicitly stated part of the verification boundary.

7.3 Permanent standard tails and unused data

7.3.1 An all-future phase test

Let $B(k)$ be a program's complete endpoint field. For a proposed standard word, write $q_i(k)$ for its 104 phase reads and d for its diagonal displacement. At phase i, epoch n, its actual read bit is

$$\mathbf{1}_{B(k)}(q_i(k) + nd) \oplus \#\{t : q_t - q_i = sd,\ s > 0 \text{ or } (s = 0, t < i),\ s \leq n\} \pmod 2. \tag{16}$$

This is obtained directly by counting all preceding planned flips. Because $B(k)$ uses finitely many rows independently of k and $d_y = \pm 2$, a phase ray meets each such row at only one possible nonnegative epoch. Check those epochs and their successors, all prior-write thresholds, zero and one epoch beyond the last change. Between them the bit is constant. At each chosen epoch the remaining condition is an affine test in k handled by Lemma 16. This proves all epochs, not a fixed number of repetitions.

There are 4,415 such all-parameter phase/epoch checks. The certified word is a cyclic phase of the standard primitive word. E0–E2 and W1–W3 finish at $(n+3, 7)$, N; W4–W6 finish at $(14, 14)$, W. Their tail drift is $(-2, 2)$. The standard-word identity is also independently checked against a complete blank-orbit state and the same infinite criterion.

Lemma 17 (Earliest entrance)**.** *Each listed completion time is the earliest permanent standard time.*

Proof. The last actual pre-completion letter differs from the letter immediately preceding the certified standard phase. An earlier permanent standard suffix would have the same unique 104-periodic backwards extension, since the standard word is primitive and both futures agree. This contradicts that letter. The argument also excludes an earlier permanent suffix during the counter phase. □

7.3.2 Protection of arbitrary extra counter lengths

E2 proves the base $l = r+2$. Increasing l prepends more copies of $\mathcal{J}(V)$. After aligning the common part, the difference lies in $x < 0$, rows $0, \dots, 4$. All finite-program reads in those rows have $x \geq 0$. For each variable scan, it is enough to check both affine endpoints; all future phase-ray encounters with those rows are checked as well. Consequently the whole actual future ignores that difference, by induction up to a hypothetical first differing read. The added cells stay in the field.

Similarly W6 extends from $r = l+6$ to every larger r: the extra V blocks are beyond $x = 4l+21$ in the five data rows, and no finite-program read or future phase-ray encounter reaches them. The final finite base in each row of Table 2 has the corresponding guard. Some small cases eventually drift right, but their rays leave the data rows before reaching the extensions; a drift sign alone is not used as a protection proof.

The two balanced exceptions and all 57 left-exhausted finite bases are replayed from (12). Their complete fields satisfy (16) and Lemma 17; the designated nine bases also satisfy the extension guards. These domains exhaust $\mathbb{N}^2$. Together with the nine programs, this proves Theorem 15.

Corollary 18 (Exact trace clipping)**.** *For $l \geq r$, replace l by $\min(l, r+2)$. For $l < r$, replace r by $\min(r, l + D(l))$, where $D(l)$ is the length of row l in Table 2 for $l < 9$, and $D(l) = 6$ otherwise. After aligning their initial heads, the original and clipped inputs have the same entire forward head and turn trace. Their complete fields differ by the static untouched extra columns. In particular $D(l) \leq 10$ for every l.*

Proof. During the initial counter exchanges the six kernels cannot reach the omitted blocks: both counters make only the shorter number of successful exchanges, with one additional right success when the left counter exhausts first. The complete post-exit states therefore differ only in those blocks. The finite and infinite guards just proved continue the equality forever. □

The same first-disagreement argument allows arbitrary finite payloads in the protected omitted half-strips, rather than only the indicated counter pattern. This is a proved guard for these specific common sources, not a procedure for discarding arbitrary cells by assumed future liveness.

7.4 Exact extrema and nonlinear entrance times

Theorem 19 (Worst entrance at fixed smaller counter)**.** *The quantity $M(m) = \max_{\min(l,r)=m} \tau(l, r)$ is finite. The maximum allows unbounded width and aspect ratio. For $m = 0, \dots, 8$, respectively, its values are*

$$17154, 6554, 2161, 2365, 4363, 8246, 5351, 5687, 7181,$$

and for every $m \geq 9$,

$$M(m) = 8m^2 + 244m + 3726. \tag{17}$$

Consequently, for all $l, r \geq 0$ and $m = \min(l, r)$,

$$\tau(l, r) \leq 8m^2 + 244m + 17154. \tag{18}$$

The coefficients $8, 244, 17154$ are lexicographically optimal among bounds of this polynomial form.

Proof. For $m \geq 9$, the choice $(l, r) = (m, m+5)$ gives (17). Subtracting the other affine delays in Table 1 proves that none is larger. For $m < 9$, maximize the three right-exit cases and the finite row

of left-exit values; the last row value already covers all larger aspect ratios. This gives the displayed table and (18). The ray $(m, m+5)$ forces first the quadratic coefficient to be at least eight, then, with eight fixed, the linear coefficient to be at least 244. With both fixed, $(l, r) = (0, 1)$ forces the constant to be at least 17154. □

The lower bound $E(l, r)$ and (18) show, uniformly in the larger counter,

$$\tau(l, r) = 8 \min(l, r)^2 + O(\min(l, r) + 1).$$

This uniformity is a dynamical conclusion: adding an arbitrarily long memory tail does not necessarily prolong the execution, because a verified finite margin screens it from every future read.

Corollary 20 (Nonsemilinear entrance graph)**.** *The complete source family* (10) *is Presburger-definable with finite fibers. Every member lies in the standard-highway basin, but the graph* $\{(l, r, \tau(l, r)) : l, r \in \mathbb{N}\}$ *is not semilinear.*

Proof. The column boundaries are affine and the two-column words use congruences modulo two, so the source family is Presburger-definable. If the entrance graph were semilinear, intersecting with $l = r = k \geq 2$ and projecting would give a semilinear graph for (11). An infinite linear component of such a function graph must contain an arithmetic ray with positive increment a in k. Its second difference in the output is zero, whereas the quadratic has second difference $16a^2 > 0$. Hence no infinite linear component is possible, and finitely many finite components cannot cover the graph. □

Every nonnegative parameter pair belongs to the highway basin, and its entrance time is evaluable by the displayed integer formulas in polynomial bit time. The correspondence between Presburger-definable sets and semilinear sets is the classical theorem of Ginsburg and Spanier [3, Theorem 1.3].

8 Random access to the complete evolving field

The formulas above describe more than the time of an event. They give a succinct representation of the entire field at any requested time. A finite arithmetic segment $[x, y, d_x, d_y, n]$ denotes $\{(x + id_x, y + id_y) : 0 \leq i < n\}$. Combining segments by XOR means that a cell is black exactly when an odd number of the segments contain it.

Theorem 21 (Exact complete-state random access)**.** *For all nonnegative binary integers* l, r, T*, the complete state* $F^T(X(l, r))$ *is computable in polynomial bit time in* $\log(l + r + T + 2)$*. Its field is an XOR of at most 20,000 finite segments. Every nonsingleton has* $d_x = 2$ *and* $d_y \in \{-2, 0, 2\}$*; coordinates and lengths have* $O(\log(l + r + T + 2))$ *bits. The representation gives the exact pose, next turn, cell membership and total black mass, and supports exact equality of two complete fields. With* M *explicitly supplied initial toggles inside the certified unread half-strips, the bound is* $20,000 + M$ *and time is polynomial in total input length.*

Here F is the ordinary one-update map. The input is the compact pair of counter lengths, and the output is a succinct complete field. We use the original coordinate frame of Section 7 throughout.

8.1 Powered words retain every write

Suppose a certified word w restores heading, has displacement d, and has phase read positions q_i relative to its entry head p. For n full copies,

$$p_{\mathrm{new}} = p + nd, \qquad B_{\mathrm{new}} = B \triangle \bigtriangleup_{i<|w|} \{p + q_i + jd : 0 \le j < n\}. \tag{19}$$

Every update flips exactly its read cell, so this identity preserves all self-intersections and earlier colors with the correct parity. A partial copy is handled by quotient and remainder and its finite list of reads. The counter and recovery words have displacement $(\pm 2, 0)$; the standard tail has diagonal displacement with components ± 2. Reversing a negatively directed finite segment normalizes d_x to 2. Singletons can use direction $(2, 0)$.

8.2 Locating the requested time

Let $m = \min(l, r)$. Before first exit, the largest completed round is

$$j = \min\left(m, \left\lfloor \frac{\mathrm{isqrt}(\lfloor T/2 \rfloor + 49) - 7}{2} \right\rfloor\right).$$

Indeed $t_j = 8j^2 + 56j = 2(2j + 7)^2 - 98$. The integer square root therefore inverts the exact round inequality without approximation. Write the complete checkpoint $\mathcal{J}((5)\mathcal{R}(j, l-j))\mathcal{R}(j, r-j)$ directly and apply (19) to at most one remaining pair of excursions. With $P = \mathtt{RLLR}$ and the kernel words K, E of Section 6, the right success is $P^j K P^j$ and its terminal call is $P^j E$. The left words are the letterwise R/L swaps of $\mathtt{LR}P^j K P^j\mathtt{RL}$ and $\mathtt{LR}P^j E$, respectively. Their execution follows from the proved reflection/complement identities.

For T after first exit, start directly with the complete exit formula (12). Scan the fixed list of recovery blocks, subtracting lengths arithmetically until the requested block is found. Its partial execution uses (19). The nine infinite parameter domains are covered by the exact affine verifiers. The 59 exceptional clipped sources have bounded parameters and recovery words of length at most 17,106, so their words may be computed by a fixed finite execution. Once $T \ge \tau$, divide $T - \tau$ by 104. The resulting finite diagonal segments describe every written cell of the whole growing wake, together with the remainder of the current period.

8.3 Restoring the original unbounded counter

Let (l_c, r_c) be the clipped pair in Corollary 18, and set $s = (2(l - l_c), 0)$. If B_0 is the complete initial source, put

$$D = B_0(l, r) \triangle (B_0(l_c, r_c) + s).$$

Every point of D is unread for the entire common future by the proved protection guards. First-disagreement induction gives

$$p_T(l, r) = p_T(l_c, r_c) + s, \qquad B_T(l, r) = (B_T(l_c, r_c) + s) \triangle D.$$

Both source fields in D are constant-sized arithmetic expressions. Thus the extra counter remains present in the output. Explicit toggles in the certified unread half-strips append singleton terms; their guard is checked at input.

8.4 Uniform size and exact set operations

Each source, checkpoint or exit field has fewer than 15 blocks of width at most four and at most five black cells per column. A repeated block has width two, so fewer than 300 segments suffice for each complete such field. The remaining counter round has phase budget below 150; the largest generic recovery program has budget 4,074, plus at most 19 remainder reads. A standard tail adds at most $104 + 103 = 207$ segments. The largest exceptional branch is bounded by

$$300 + 17,106 + 207 + 2 \cdot 300 = 18,213 < 20,000.$$

The last two terms restore the source difference. Cancellation only reduces the count. Coordinates have magnitude $O(l + r + T + 1)$ by the source formulas and unit-step motion. The algorithm uses a bounded number of integer operations, including one square root, on numbers of logarithmic bit length. This proves the representation and bit-complexity claims.

Membership is the parity of divisibility and interval tests on the segments. For total mass, group segments by direction d_y, x parity a, and intercept $y - d_y(x - a)/2$. XOR-sweep their integer endpoints, obtaining disjoint intervals on each line. Distinct parallel lines are disjoint. Two nonparallel lines intersect in at most one point; exact integer determinants and interval tests identify every such intersection. Group coincident points and collect all incident segment identifiers, so triple intersections are counted together. If $m(p)$ is the number of retained segments meeting at an intersection, then

$$|B| = \sum_i |S_i| - \sum_p (m(p) - (m(p) \bmod 2)).$$

Ordinary points occur once; this correction replaces each crossing multiplicity by its XOR parity. There are polynomially many endpoints and pair intersections in the number of terms, regardless of numeric segment length. Applying the same mass calculation to the XOR of two fields decides equality. Membership at the returned pose gives the next turn. These operations complete the proof of Theorem 21. □

9 Computational proof obligations

The finite certificate has four roles: it verifies the response products and ranks of Proposition 3, the local products of Proposition 4, the complete recurrence graph of Lemma 5, and the integer inequalities and tight edges of Lemma 6. Each finite check is connected above to an induction covering all widths. Appendix A describes the verifier, reconstruction and literal trajectory controls.

10 Consequences

The response catalogue resolves both the linear upper bound and the exact maximum for four rows. It also has an infinite-strip consequence. If an orbit stays forever in a four-row band, each fixed column is updated at most 34 times: every finite prefix fits inside a finite rectangle, to which Proposition 4 applies. Such a confined orbit is therefore transient: every fixed cell is visited only finitely often. This conclusion also holds on an infinite initial colouring.

11 Limitations

The infinite-strip consequence assumes that the orbit stays in the band. A general planar orbit can leave a band and return after modifying its exterior. Extending the method requires interfaces that account for those excursions. The five-row family supplies a quadratic lower bound; the exact maximal five-row escape function remains to be determined. Its recovery and succinct state algorithm apply to the explicitly encoded two-counter family. The 20,000-term bound is conservative, and explicit enumeration of every black coordinate has output cost proportional to the field being printed. The finite tables and their coverage arguments are verified by exact programs and written inductions. Their translation into a formal proof assistant remains a separate task; the computational trust base is described in Appendix A.

A Computational verification and reproduction

A.1 Certificate architecture and controls

The supplied code has a Python constructor and a separately written C++17 verifier. The latter reconstructs all column products from its own literal turn/flip rule, checks the response ranks, independently builds the expected edge list at every weighted state, checks completeness of the goal set, and recomputes all dynamic values. It then verifies every inequality and tight-edge obligation. A second Python integer DP repeats the affine-vector calculation and the induction check.

The canonical response/graph serialization has SHA256

```
3e01b0422cf2d921fb0ed4aaa690d8f89ffacfa92973a25c3aea590ac8d1110b
```

and the checked A, B, d, e row serialization has SHA256

```
4fb5165094326189694eb227157b578497e089f19eff2abed050b14286c27bca
```

Readable deterministic reconstruction avoids a large opaque table dump. Hashes pin the exact finite claims; the mathematical verification is the local composition, coverage and integer conditions, not the hash itself.

The scoped verification includes the following distinct controls.

- All 1,076 side/parity-specific representatives have their complete future same-side reply trees compared to direct full-colour simulation. Every colouring through width three is checked as well: 17,472 side/parity behaviours.
- Native literal simulation exhausts 4,399,360 colouring/pose inputs through width four, making 20,978,524 updates. It matches the exact maxima but is not the proof for larger widths.
- Twenty-nine reconstructed maximizing seeds, through width 1001, are replayed independently in Python, totaling 89,950 updates. Additional random wide cases and full-plane exterior-marker controls preserve complete fields.
- Altered response data, missing coverage counts, an incorrect recurrence weight and a damaged affine vector are rejected. Resource exhaustion never means a negative answer or an accepted certificate.

The default check runs in a disposable directory and compares a fresh report to a frozen report without overwriting it. Generation is a separate explicit mode. The finite tables and their coverage arguments have not been translated into a formal proof assistant. The trusted base includes the written reductions, the exact verifier source, language implementations and integer execution.

From the repository root run

```
python3 four_row_escape/reproduce.py
```

The only runtime requirements are Python 3.10 or later and a C++17 compiler. The implementation is in `four_row_escape/verification/four_row_boundary/`. The source archive contains the full verifier, the reconstruction code, and the expected report used to check reproduction. These computational sources are also fixed at repository commit `e932522`, four-row project; `sources.json` lists their individual hashes.

An optional witness export is produced by

```
python3 four_row_escape/verification/four_row_boundary/check.py \
  --witnesses /tmp/four-row-maximizers.txt
```

Each row is `n x y h time mask0 ... mask(n-1)`. A mask's bit y is the initial black colour at that row; all other cells are white unless expressly added as exterior controls. The first-exit calculation does not read outside the rectangle. The source also records an attaining 34-update column and an autonomous 21-crossing cut witness.

A.2 Five-row, recovery and complete-state certificates

The six literal kernels and all-parameter round induction in Section 6 establish the autonomous construction. Its dedicated checker also reconstructs the finite height-one through height-four response catalogues. Python and C++ full-field controls cover both exhaustion orders and all zero-counter boundaries.

For recovery, the source package contains the nine literal affine-word programs, the 59 base cases, their complete field compiler and two exact integer epoch checkers. Each infinite parameter ray and final infinite epoch interval is covered by the finite sign and intersection reductions proved in Section 7. The extra-counter guards are checked on both finite programs and every possible tail encounter with the data rows.

The random-access checker compares 6,291 intermediate complete states on 111 original sources with literal implementations. It checks finite segment arrangements against direct sets and tests exact one-update identities on parameters as large as 10^{120}. These are controls on the implementation; the uniform term and bit bounds are proved in Section 8. All three components are run by the same reproduction entrypoint. Individual modules can also be checked with `-component five`, `-component basin`, or `-component access`.

Assistance. AI-assisted tools supported proof development, exposition and verification code. The written coverage arguments, immutable witnesses and exact verifier programs specify the mathematical evidence.

References

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